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Advancing Petrophysical Property Estimation through Geophysical Techniques

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Abstract

This study focuses on developing field-applicable equations that translate spectral induced polarization (SIP) signals to petrophysical properties, with experimental insights gained from laboratory measurements. The objective is to advance non-invasive methods for estimating key properties, such as porosity and permeability, across various rock types, which are applicable in hydrogeology, mineral and fossil fuel exploration, and management. The study involves controlled laboratory measurements using SIP and conventional petrophysical methods on core samples representing a range of typical subsurface media. These measurements are compared to known petrophysical properties to calibrate and assess the reliability of SIP-based equations. A specific focus is on examining the combined use of SIP and traditional petrophysical measurements, aiming to refine existing empirical relationships for porosity and permeability estimations, and assessing their applicability across various rock types. Laboratory measurements reveal that SIP effectively captures variations across diverse rock types, indicating a robust potential for broader applicability. However, permeability estimations remain challenging due to the influence of complex pore structures. Preliminary results show that incorporating traditional petrophysical methods enhances the accuracy of permeability estimates in heterogeneous rocks. The synergy between SIP and other petrophysical measurements provides a more comprehensive view of subsurface properties, highlighting how integrated measurements can improve property estimation in laboratory settings.

Introduction

Deriving petrophysical parameters from measured signals has long been the ultimate goal of multiple geophysical sub-disciplines, including environmental and engineering geophysics, as well as hydrogeophysics. Extensive research in this direction has been conducted in the last decade, with promising results (Revil et al., 2010; Robinson et al., 2018; Weller et al., 2015, 2016, 2019; Zhang et al. 2012).

Great strides have been made in permeability prediction using complex conductivity, also known as spectral-induced polarization (SIP). Unlike conventional resistivity methods, SIP captures both the real and imaginary components of conductivity, providing insights into the capacitive behavior of subsurface materials. This frequency-dependent response is sensitive to variations in pore geometry, grain surface properties, and fluid conductivity, making SIP particularly suited for characterizing pore-scale processes and estimating hydraulic properties such as permeability and porosity. The method's ability to resolve subtle subsurface features has led to its growing application in hydrogeophysics, environmental monitoring, and reservoir characterization, where traditional measurements may fall short in capturing heterogeneities. [Robinson et al. \(2018\)](#) summarized the efforts and compared standard geometrical length scale approaches with geophysical ones based on SIP ([Fig. 1](#)).

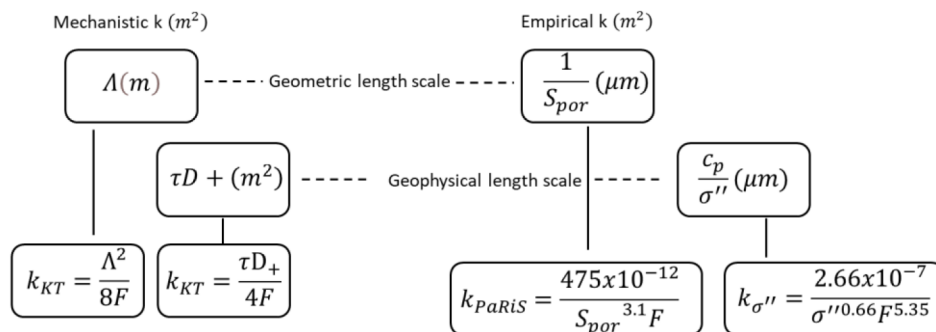


Figure 1—Summarized overview of geometric and geophysical scales permeability estimation (modified from Robinson et al., 2018).

The important finding from this research ([Robinson et al., 2018](#)) is that estimating the order of magnitude of permeability is possible using SIP data with some exceptions ([Fig. 2](#)). However, the work also identified current limitations and concerns, such as the dependence of the estimation on the formation factor that cannot be accurately extracted from field measurements. It should be mentioned that this limitation also applies to geometric length scale approaches. In addition, mineralogy and diffusion co-efficient variability can cause issues with the estimation. In a follow-up work, [Weller and Slater \(2019\)](#) attempted to address these issues, improving the outcomes and suggesting further advancements.

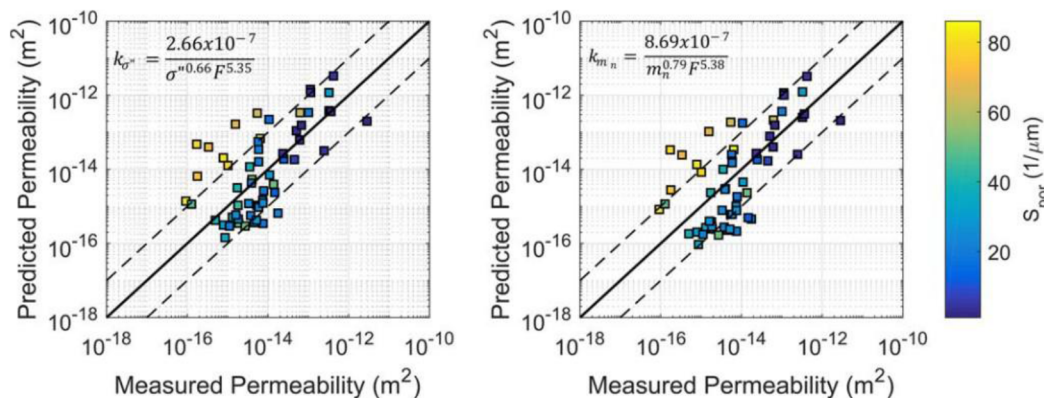


Figure 2—Permeability predicted based on SIP data ([Robinson et al., 2018](#)).

Moreover, laboratory research showed that SIP measurements can accurately estimate the formation factor (F) of a sample without the need for time-consuming multi-salinity measurements currently used ([Osterman et al., 2016](#)). The electrical conductivity c and hydraulic conductivity K are expected to have a relationship since they have a common behavior in earth media: water and electrical current both flow

through the interconnected pore space. Despite this, there is no universal equation to translate electrical properties to K. Slater (2007) reviewed all approaches and concluded that resistivity alone cannot be used to estimate K, at least not universally and suggested that IP is probably a better approach. Revil et al. (2010) proposed an equation to estimate K from SIP data, and Slater et al. (2014) successfully applied it in lab sediment experiments. The effective porosity is linked to both permeability and hydraulic conductivity and could be estimated by a variety of equations, or when tortuosity T is known, from the following equations:

$$F = \frac{T}{\phi} \quad (1)$$

$$K = \frac{r^2 \cdot \phi}{8 \cdot T} \quad (2)$$

where F is formation factor, T is tortuosity, ϕ is porosity (effective in this case), K is hydraulic conductivity, and r is pore radius. Factors F, K, and r can be estimated using SIP.

Schwarz (1962) described the relationship between electrical relaxation time ϕ (which is the reciprocal of the peak frequency of the SIP signal) and particle radius. The sensitivity of SIP signals to grain size has been extensively studied. Wong (1979) for example, developed a model to describe the SIP response of metal particles showing that the volume of the target affects the magnitude of the response, while the grain size is the peak frequency. The model has been successfully used in metal sulfide studies (Placencia-Gomez, 2015). The same behavior, frequency dependence with grain size, is also observed in earth samples without metallic particles. While this relationship is not disputed, the current discussion on the signal source for SIP; is the grain size, the pore size, or a combination of the two. Recent research indicates that in the absence of metallic particles, the pore size dictates the signal, while in the presence of metals, the metallic particle size is the contributing factor. While this is an interesting research question, the important aspect is that SIP can provide the information needed to describe grain or pore size distribution. In recent work, Kruschwitz et al. (2020) utilized SIP, along with other standard methods (MIP, NMR, g-CT), showing that it is indeed a powerful tool for pore characterization, but more importantly, showing that there is further potential to the method once the dominant signal sources and processes are better understood and described.

The primary purpose of this study is to evaluate the effectiveness and limitations of using SIP as a non-invasive proxy for porosity and permeability estimation across diverse lithologies. By correlating SIP-derived parameters with conventional laboratory measurements, the study aims to refine empirical models that can be applied in practical reservoir characterization workflows. Emphasis is placed on assessing how well SIP can capture the influence of microstructural variations, such as pore size and connectivity, on fluid flow properties. Ultimately, the goal is to establish a more accurate and scalable approach to petrophysical property estimation that supports improved decision-making in subsurface exploration, water resource management, and hydrocarbon recovery.

Methods

Specific core holders (Fig. 3.) with adjustable lengths were manufactured to measure the SIP responses of core specimens. This adaptability is crucial for accommodating core specimens of varying sizes, ensuring accurate and consistent measurements across different sample lengths. However, before utilizing this core holder for SIP measurements, it is essential to determine the geometric factor of the system. The geometric factor plays a critical role in interpreting SIP responses as it accounts for the effects of the core holder's geometry on the measured data. This factor must be accurately calculated to ensure that the SIP measurements reflect the true properties of a core specimen rather than artifacts introduced by the measurement setup. Determining the geometric factor involves carefully analyzing the core holder's dimensions, the positioning of the electrodes, and the configuration of the measurement apparatus.

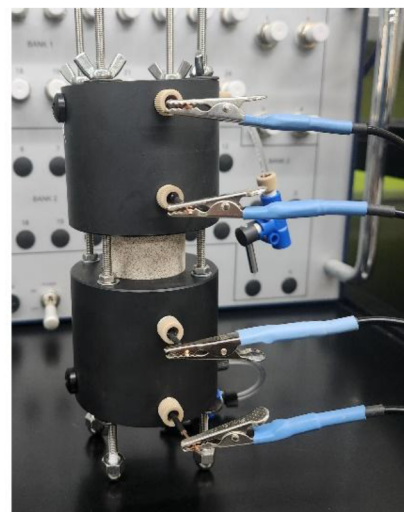
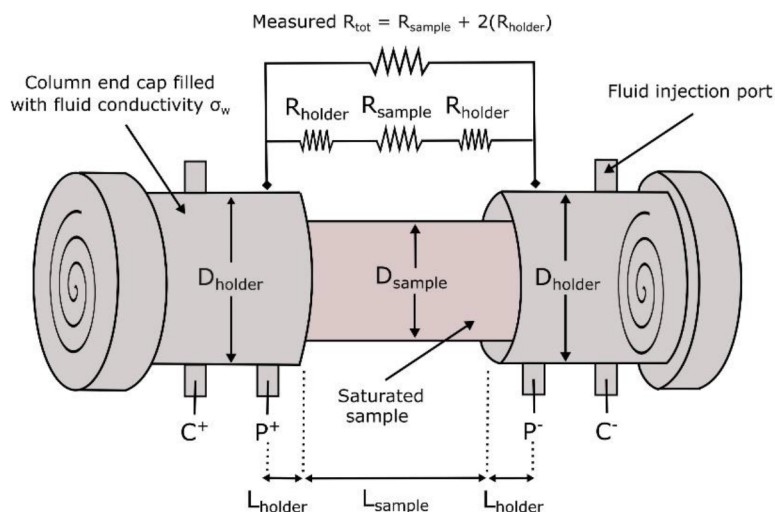


Figure 3—Setup used for electrical resistivity measurements using a specimen holder.

To ensure that subsequent data analysis is based on representative measurements, the electrical resistivity must be determined with high precision. This is achieved by injecting current through the sample using current electrodes and recording the resulting voltage between potential electrodes, which yields electrical resistance via the relation $R = AV/I$. The resistivity or conductivity is then calculated using a geometric factor k , which depends on the physical setup of the core holder. When electrodes are not in direct contact with the sample, additional resistance from the endcaps or holder must be corrected. This is typically done by calibrating the setup with fluids of known conductivity, plotting measured resistance against known resistivity to derive the geometric factor. In complex configurations where one-dimensional current flow cannot be assumed, numerical modeling tools such as COMSOL Multiphysics are employed to simulate current distribution and refine the geometric correction. These steps are crucial to ensure that the final SIP-derived parameters accurately represent the intrinsic properties of the specimen, thereby supporting a reliable petrophysical interpretation.

Although a variety of rock types (carbonate, sandstone, and limestone) were tested across a wide range of porosity and permeability values, as illustrated in Fig. 4, only the results from the carbonate samples will be presented in detail for simplicity. These samples provide a representative dataset while reducing complexity in the analysis and interpretation. The measurements conducted include SIP, Nuclear Magnetic Resonance (NMR) for porosity evaluation, and liquid permeability flow tests. SIP measurements were performed over a frequency range from 0.1 Hz to 10,000 Hz, capturing both amplitude and phase shift values with high resolution (six measurements per logarithmic decade). These data were acquired using the PSIP system developed by Ontash & Ermac (<https://www.ontash.com/>), a laboratory-optimized SIP instrument known for its precision and stability.

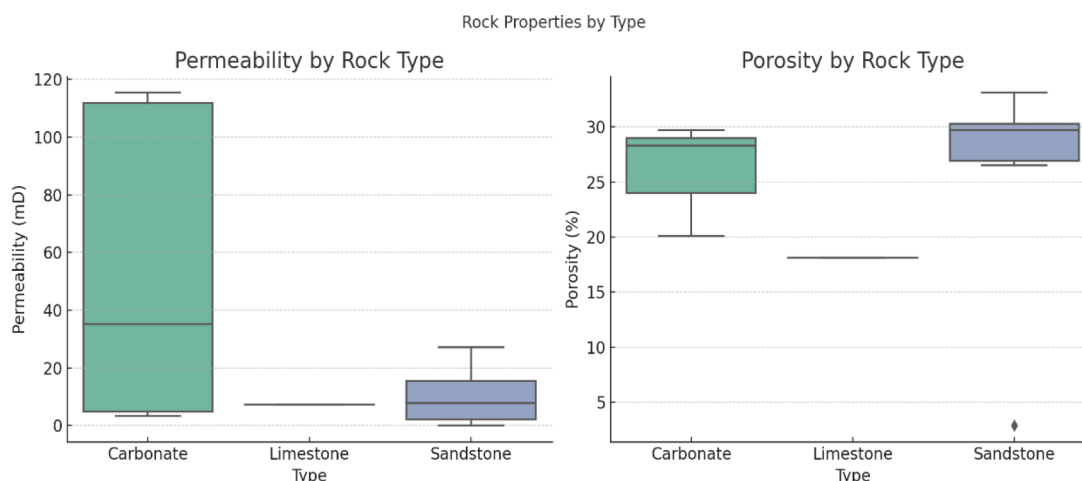


Figure 4—Distribution of median values and variability of permeability (mD) and porosity (%) for carbonate, limestone, and sandstone samples.

Results

Based on the NMR and liquid permeability measurements, the results are summarized in [Table 1](#). Each core sample displays a distinct T_2 relaxation time distribution, which reflects differences in pore geometry, fluid retention capacity, and lithological composition. Sample 01, with a total NMR porosity of 28.3 %, shows a well-developed pore network, closely resembling Sample 02, which has a slightly higher porosity of 29.0 %. Sample 03 exhibits the highest porosity at 29.7 % and features a well-defined peak at longer T_2 relaxation times, indicating the presence of larger, more connected pores that facilitate higher fluid mobility. In contrast, Sample 04, with a porosity of 20.1 %, shows a dominant peak at shorter T_2 values, suggesting a pore system composed primarily of smaller pores and bound water. Sample 05, with a total porosity of 24.0 %, displays a bimodal T_2 distribution, indicative of a heterogeneous pore structure with both fine and coarse pores.

Table 1—Rock samples presented, including rock type, permeability and porosity.

S.N.	ID	Type	Permeability (mD)	Porosity (%)
1	C1	Carbonate	35.225	28.3
2	C2	Carbonate	115.381	29.0
3	C3	Carbonate	111.8214	29.7
4	C4	Carbonate	4.7951	20.1
5	C5	Carbonate	3.3289	24.0
6	L1	Limestone	7.2837	18.1

[Fig. 5](#) shows the SIP results of six core samples (1 to 6). In the phase shift plot ([Fig. 5a](#)), most samples display an increasing trend with frequency, particularly beyond 100 Hz, indicating enhanced polarization effects at higher frequencies. The inset plot focuses on the 11000 Hz range, where subtle differences become evident and help to differentiate each sample's early-stage frequency response. The impedance plot ([Fig. 5b](#)) shows that all samples have a stable response across the frequency spectrum, with no major frequency-dependent trends. Sample C6 has the highest impedance, consistent with lower real conductivity, whereas sample C3 has the lowest impedance, aligning with its high phase response and higher conductivity. Real conductivity is frequency-independent for all samples, reflecting the purely resistive (ohmic) component. The imaginary conductivity ([Fig. 5d](#)) shows a frequency-dependent increase, echoing the phase shift trends.

This component demonstrates capacitive behavior associated with polarization. The insert zoom highlights early- frequency responses, which are critical for interpreting subtle polarization mechanisms...

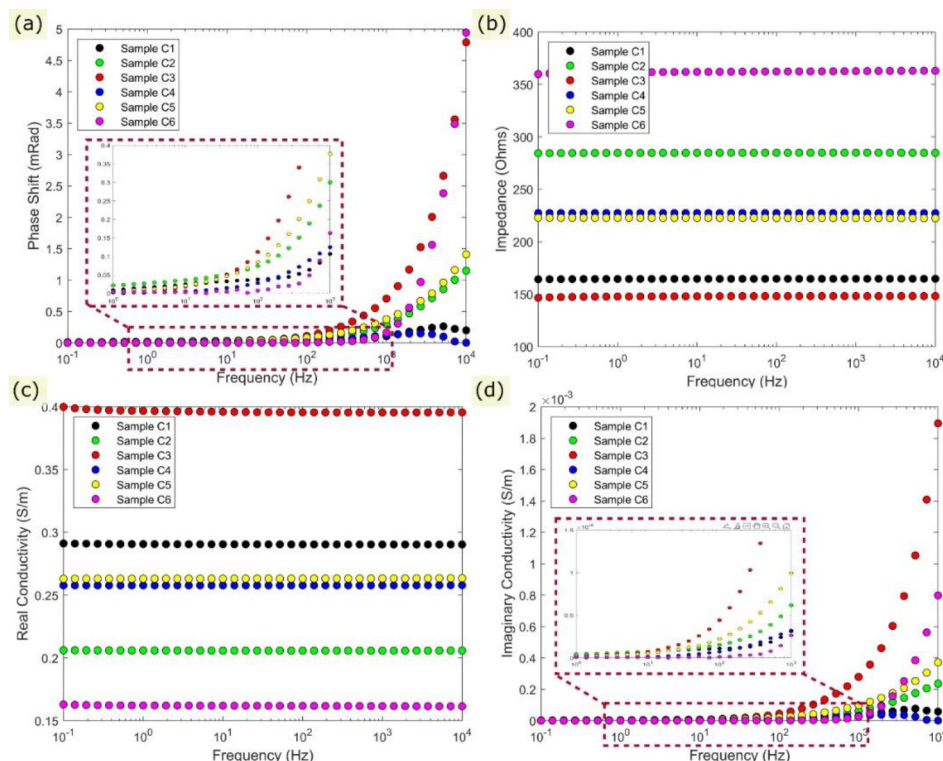


Figure 5—SIP results of samples 1 to 6. (a) Phase shift (in mRad) vs frequency; insert zoom into the 1-1000 Hz, (b) Impedance (in Ohms), (c) Real conductivity (in S/m), and (d) Imaginary conductivity (in S/m); insert zoom into the 1-1000 Hz.

Fig. 6 indicates that higher porosity tends to be associated with stronger polarization effects, as captured by the imaginary conductivity. This supports the interpretation that pore structure plays a key role in the SIP response of core samples.

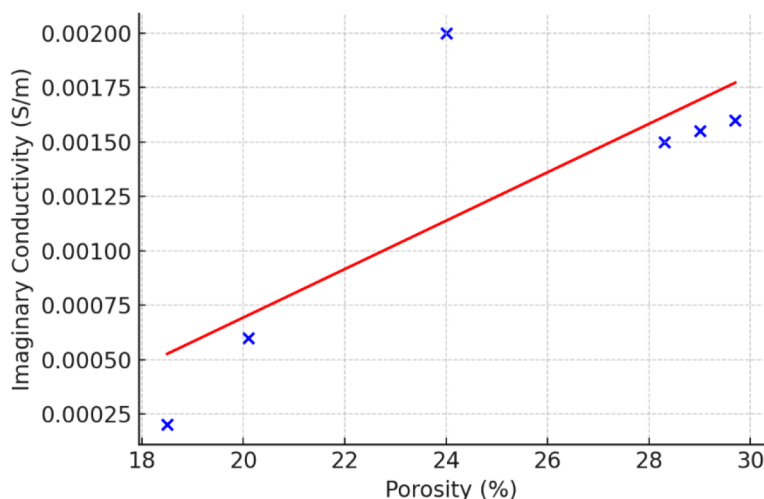


Figure 6—Relationship between imaginary conductivity at 1 kHz and porosity with a linear fit applied to the data. The red line represents the best-fit linear regression

Fig. 7 presents a comparative analysis between porosity and permeability estimates derived from SIP measurements and those obtained from conventional core analysis methods. In Fig. 7a, porosity estimates

were computed using an empirical log-log relationship calibrated from the real component of complex conductivity (σ') at 100 Hz. These SIP-derived porosities are compared against porosity values measured via NMR, which serves as a benchmark due to its sensitivity to pore volume. The plot includes a 1:1 line and sample identifiers (1-6) to highlight deviations and illustrate the degree of agreement. The results demonstrate that SIP, when properly calibrated, can provide reliable porosity estimates, particularly in carbonate settings. Fig. 7b shows the comparison between SIP-estimated permeability, calculated from the imaginary conductivity (σ'') at 100 Hz using a power-law relationship, and laboratory-measured permeability from core flooding experiments. While the match is good for mid- to high-permeability samples, deviations are more evident in the lower-permeability range due to limitations in resolution and pore-throat connectivity effects. The 1:1 line provides a reference for visualizing estimation accuracy. Together, the results confirm that SIP can serve as a valuable, non-invasive proxy for petrophysical characterization when interpreted appropriately.

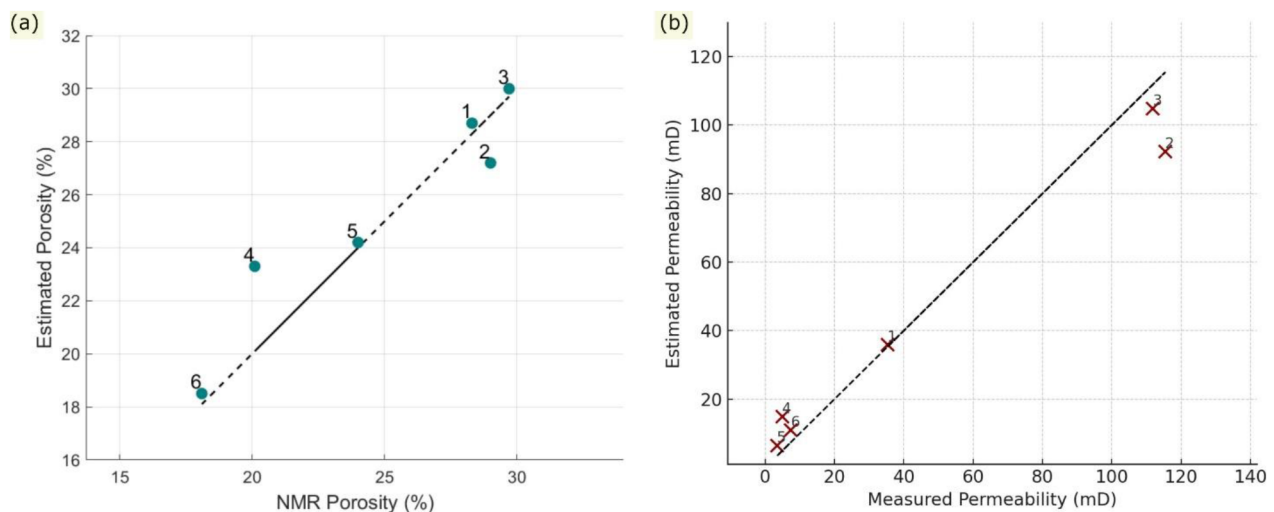


Figure 7—Comparison between laboratory-derived and SIP-estimated properties. (a) NMR porosity versus porosity estimated from SIP parameters, showing a strong correlation with a segmented linear trend. (b) Measured core permeability versus permeability estimated using imaginary conductivity at 100 Hz, plotted against the 1:1 reference line.

This study demonstrated the practical potential of SIP as a non-invasive technique for estimating fundamental petrophysical properties, specifically porosity and permeability, across a range of carbonate core samples. Through careful laboratory experimentation and comparative analysis against benchmark methods such as NMR and core-flooding permeability testing, the results support SIP as a valuable proxy tool for petrophysical characterization when properly calibrated and interpreted. The SIP measurements conducted over a broad frequency range revealed consistent, frequency-dependent behaviors in both the phase response and imaginary conductivity (σ''), which were particularly sensitive to variations in pore structure, connectivity, and fluid-rock interaction. The real conductivity (σ'), which remained largely frequency-independent, provided the basis for estimating porosity using a log-log empirical relationship. Meanwhile, permeability estimations were derived from σ'' at 100 Hz using an approach inspired by prior models in the literature. In terms of porosity, the SIP-derived estimates aligned well with NMR measurements across most samples, as shown in Fig. 7a. The inclusion of a 1:1 reference line highlighted that, although minor deviations existed, particularly in lower-porosity samples, the general trend followed a predictable and consistent pattern. This consistency highlights the ability of SIP, particularly the resistive (ohmic) component, to serve as a viable complement or alternative to NMR, especially in scenarios where traditional laboratory analysis is not feasible or where core data are limited. The permeability comparison (Fig. 7b) also highlighted significant insights. For samples with moderate to high permeability, SIP estimates were in reasonable agreement with laboratory values. However, the method's resolution decreased notably

in low-permeability settings. This finding is in line with known limitations of SIP, where polarization mechanisms are less active or less detectable in pore systems dominated by bound fluids or micropores. Despite these discrepancies, the general trends followed the expected behavior, reinforcing the relationship between imaginary conductivity and flow-related parameters.

Conclusions

One of the key conclusions is that SIP can reliably detect changes in pore structure that affect both porosity and permeability. The imaginary conductivity response in particular serves as an indicator of pore connectivity and capacitive storage, both of which are tied to hydraulic behavior. For instance, samples C1, C2, and C3, which had high porosity and permeability, exhibited strong SIP signals with characteristic frequency-dependent trends, while samples C4 and C5, which had much lower permeability, showed attenuated responses. From a methodological perspective, the success of this study relied on carefully designed core holders and calibration procedures that accounted for geometric factors and contact resistances. Without accurate geometric factor corrections, SIP measurements risk misrepresenting the intrinsic electrical behavior of the rock. In terms of broader implications, the findings support an expanded role for SIP in integrated petrophysical workflows. When combined with NMR, permeability tests, and potentially other imaging tools like micro-CT or MIP (mercury intrusion porosimetry), SIP adds an important electrical dimension that captures pore-scale processes otherwise overlooked by mechanical or purely resistive methods. This multi-physics approach is especially valuable in unconventional or carbonate reservoirs, where porosity and permeability are governed by a combination of primary deposition and diagenetic modification, resulting in complex pore systems that are challenging to characterize using a single method. However, this study also underscores several limitations. First, SIP's effectiveness is reduced in formations with very low electrical conductivity or poorly developed polarization effects. Additionally, the empirical relationships used to estimate petrophysical properties are inherently site- and lithology-specific. The models developed here for carbonate rocks may not directly translate to sandstones, shales, or unconsolidated sediments without recalibration. Second, the presence of metallic inclusions or conductive minerals can complicate the interpretation of SIP data, requiring mineralogical corrections or more advanced modeling approaches.

Looking ahead, future work should focus on refining the frequency-specific interpretations of SIP signals to isolate more accurately the contributions from pore size distribution, mineral interfaces, and saturation state. Developing universal or semi-empirical models that incorporate tortuosity, formation factor, and surface conductivity could also enhance SIP's predictive capabilities. Moreover, field-scale validation, especially in borehole or surface-based applications, is essential to demonstrate the scalability of these laboratory findings. Integrating SIP into time-lapse monitoring workflows or reservoir simulation models could further extend its utility beyond static characterization to dynamic reservoir management. In summary, SIP has demonstrated promising utility as a diagnostic and predictive tool for estimating petrophysical properties. While challenges remain in standardization and low-permeability resolution, its ability to capture both resistive and capacitive behavior provides a unique window into subsurface structure and flow.

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